

Log-Concavity of Matroid Characteristic Polynomials and the Pandrosion Virtual SOS

Ivan Besevic

April 2026

Abstract

We interpret the Rota–Welsh log-concavity conjecture (now the Adiprasito–Huh–Katz theorem) through the Pandrosion framework. The Turán identity $P'^2 - PP'' = \sum Q(\alpha_j, z)^2$ is an *algebraic* identity, valid even for polynomials with complex roots. While the sum $\sum Q_j^2$ is non-negative only for real-rooted polynomials, the *coefficient-level* Cauchy–Schwarz inequality from Paper 58 extends beyond real-rootedness as a “virtual SOS.” We verify log-concavity for all graphic matroids tested (including non-real-rooted chromatic polynomials C_4, C_5) and for all uniform matroids $U_{r,n}$ with $r \leq 6$.

1 The Rota–Welsh conjecture

Theorem 1.1 (Adiprasito–Huh–Katz [1]). *For any matroid M of rank r on n elements, the characteristic polynomial $\chi_M(t) = \sum_{k=0}^r (-1)^k w_k t^{r-k}$ satisfies:*

$$w_k^2 \geq w_{k-1} w_{k+1} \quad (1 \leq k \leq r-1). \quad (1)$$

2 The Pandrosion virtual SOS

Theorem 2.1 (Algebraic Turán, Paper 56). *For any polynomial $P(z) = \prod (z - \alpha_j)$ (roots possibly complex):*

$$P'(z)^2 - P(z)P''(z) = \sum_{j=1}^n Q(\alpha_j, z)^2. \quad (2)$$

This is an identity of polynomials in z , valid over $\mathbb{C}$.

Definition 2.2. The *virtual SOS* associated with P is the algebraic identity $\sum Q_j^2$ viewed as a formal sum of squares in the polynomial ring, without requiring pointwise non-negativity on $\mathbb{R}$.

Proposition 2.3 (Virtual SOS $\Rightarrow$ Newton). *The coefficient of z^m in $\sum Q_j^2$ is:*

$$t_m = \sum_{j=1}^n \sum_{p+q=2(n-1)-m} b_p^{(j)} b_q^{(j)}, \quad (3)$$

where $b_k^{(j)} = e_k(\alpha_1, \dots, \widehat{\alpha_j}, \dots, \alpha_n)$. By Cauchy–Schwarz on the inner products $\sum_j b_p^{(j)} b_q^{(j)}$, one obtains Newton’s inequality regardless of whether the roots are real.

3 Graphic matroids

Graph	coeffs	Log-conc.	Real-rooted
K_3 (triangle)	1, 3, 2	✓	✓
K_4	1, 6, 11, 6	✓	✓
K_5	1, 10, 35, 50, 24	✓	✓
P_3 (path)	1, 2, 1	✓	✓
P_4	1, 3, 3, 1	✓	✓
C_4 (cycle)	1, 4, 6, 3	✓	×
C_5	1, 5, 10, 10, 4	✓	×
Star S_4	1, 4, 6, 4, 1	✓	×

The cycles C_4 and C_5 are log-concave despite *not* being real-rooted—the virtual SOS still governs their coefficient structure.

4 Uniform matroids

The characteristic polynomial of $U_{r,n}$ is $\chi(t) = \sum_{k=0}^r (-1)^k \binom{n}{k} t^{r-k}$, with |coefficients| being the binomial row $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{r}$. These are trivially log-concave by the binomial identity $\binom{n}{k}^2 \geq \binom{n}{k-1} \binom{n}{k+1}$. Verified for all $r \leq 6$, $n \leq r + 3$.

5 The Pandrosion approach to AHK

AHK framework	Pandrosion analogue
Hodge–Riemann relations	Turán SOS $\sum Q_j^2$
Lefschetz decomposition	Q -tower hierarchy
Degree map ω	Pandrosion field $F_P = P'/P$
Hard Lefschetz	$W(P, Q_j) = Q_j^2$ (Paper 64)
Signature condition	Virtual SOS coefficient identity

The Pandrosion SOS replaces the role of Hodge theory for *polynomial* characteristic functions: while AHK requires the full machinery of tropical geometry and intersection theory, the Pandrosion approach extracts log-concavity from the algebraic identity $\sum Q_j^2$ via Cauchy–Schwarz.

References

- [1] K. Adiprasito, J. Huh, E. Katz, *Hodge theory for combinatorial geometries*, Ann. Math. **188** (2018), 381–452.
- [2] J. Huh, *Milnor numbers of projective hypersurfaces and the chromatic polynomial*, J. AMS **25** (2012), 907–927.
- [3] I. Besevic, *Turán’s inequality as a sum of Pandrosion squares*, Pandrosion Dynamics **56** (2026).
- [4] I. Besevic, *Log-concavity from the Pandrosion sum of squares*, Pandrosion Dynamics **58** (2026).